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Module 1 Assignment 1

1. Explain the interdependence between data mining and big data

~~Data~~ ^{Big} ~~mining~~ provides the infrastructure while big data ~~Data~~ mining provides the mechanism or the algorithm

2. What is Hadoop and its importance

Hadoop is the process of extracting valuable insights, patterns, and trends from massive datasets

Importance of Hadoop

- i) Used to extract meaningful data from data infrastructure
- ii) Supports ingestion and storage of data, which drives
- iii) Used in integration with machine learning algorithms

ASSIGNMENT SUBMISSION MODULE 1

Real-World Significance for Kenyan Demographics

Applying these clustering techniques to Kenyan demographic data has important implications:

- **Urban Planning:**
 - DBSCAN can better identify high-density urban areas like Nairobi and informal settlements, helping in infrastructure development.
- **Resource Allocation:**
 - K-Means can group counties with similar socio-economic profiles, enabling efficient distribution of healthcare, education, and financial services.
- **Financial Inclusion:**
 - Banks and fintech companies can use clustering to identify underserved populations and tailor mobile banking solutions.
- **Policy Making:**
 - Insights from clustering can support government strategies in poverty reduction, employment, and regional development.
- **Disaster and Risk Management:**
 - DBSCAN can highlight isolated or vulnerable populations in sparsely populated areas, improving response planning.

Conclusion

Both K-Means and DBSCAN offer valuable insights when analyzing Kenyan demographic data. K-Means is efficient and suitable for large, structured datasets, while DBSCAN provides deeper insights into complex, real-world patterns with noise and irregular distributions. A hybrid or combined approach may yield the most meaningful results for decision-making in Kenya.